

Let us conclude this section with a few comments about terminology. For some diagrams of spaces such as sequences $X_1 \rightarrow X_2 \rightarrow \cdots$ the amalgamation $\sqcup X$ can be regarded as the direct limit of the vertex spaces X_v with respect to the edge maps f_e . Following this cue, the space $\sqcup X$ is commonly called the direct limit for arbitrary diagrams, even finite ones. If one views $\sqcup X$ as a direct limit, then ΔX becomes a sort of homotopy direct limit. For reasons that are explained in the next section, direct limits are often called ‘colimits.’ This has given rise to the rather unfortunate name of ‘hocolim’ for ΔX , short for ‘homotopy colimit.’ In preference to this we have chosen the term ‘realization,’ both for its intrinsic merits and because ΔX is closely related to what is called the geometric realization of a simplicial space.

Exercises

1. Show that for a sequence of maps $X_0 \xrightarrow{f_1} X_1 \xrightarrow{f_2} \cdots$, the infinite iterated mapping cylinder $M(f_1, f_2, \cdots)$, which is the union of the finite iterated mapping cylinders $M(f_1, \cdots, f_n)$, deformation retracts onto the mapping telescope.
2. Show that if X is a complex of spaces in which all the maps are homeomorphisms, then the projection $\Delta X \rightarrow \Gamma$ is a fiber bundle.
3. What is the nerve of the cover of a simplicial complex by the open stars of its vertices? [See Lemma 2C.2.]
4. Show that Proposition 4G.2 and its corollary hold also for CW complexes and covers by families of subcomplexes. [CW complexes are paracompact; see [VBKT].]

4.H Eckmann-Hilton Duality

There is a very nice duality principle in homotopy theory, called Eckmann-Hilton duality in its more refined and systematic aspects, but which in its most basic form involves the simple idea of reversing the direction of all arrows in a given construction. For example, if in the definition of a fibration as a map satisfying the homotopy lifting property we reverse the direction of all the arrows, we obtain the dual notion of a **cofibration**. This is a map $i: A \rightarrow B$ satisfying the following property: Given $\tilde{g}_0: B \rightarrow X$ and a homotopy $g_t: A \rightarrow X$ such that $\tilde{g}_0 i = g_0$, there exists a homotopy $\tilde{g}_t: B \rightarrow X$ such that $\tilde{g}_t i = g_t$. In the special case that i is the inclusion of a subspace, this is the homotopy extension property, and the next proposition says that this is indeed the general case. So a cofibration is the same as an inclusion satisfying the homotopy extension property.

Proposition 4H.1. *If $i: A \rightarrow B$ is a cofibration, then i is injective, and in fact a homeomorphism onto its image.*

Proof: Consider the mapping cylinder M_i , the quotient of $A \times I \amalg B$ in which $(a, 1)$ is identified with $i(a)$. Let $g_t : A \rightarrow M_i$ be the homotopy mapping $a \in A$ to the image of $(a, 1 - t) \in A \times I$ in M_i , and let $\tilde{g}_0$ be the inclusion $B \hookrightarrow M_i$. The cofibration property gives $\tilde{g}_t : B \rightarrow M_i$ with $\tilde{g}_t i = g_t$. Restricting to a fixed $t > 0$, this implies i is injective since g_t is. Furthermore, since g_t is a homeomorphism onto its image $A \times \{1 - t\}$, the relation $\tilde{g}_t i = g_t$ implies that the map $g_t^{-1} \tilde{g}_t : i(A) \rightarrow A$ is a continuous inverse of $i : A \rightarrow i(A)$. $\square$

Many constructions for fibrations have analogs for cofibrations, and vice versa. For example, for an arbitrary map $f : A \rightarrow B$ the inclusion $A \hookrightarrow M_f$ is readily seen to be a cofibration, so the analog of the factorization $A \hookrightarrow E_f \rightarrow B$ of f into a homotopy equivalence followed by a fibration is the factorization $A \hookrightarrow M_f \rightarrow B$ into a cofibration followed by a homotopy equivalence. Even the definition of M_f is in some way dual to the definition of E_f , since E_f can be defined as a pullback and M_f can be defined as a dual pushout. In general, the **pushout** of maps $f : Z \rightarrow X$ and $g : Z \rightarrow Y$ is defined as the quotient of $X \amalg Y$ under the identifications $f(z) \sim g(z)$.

$$\begin{array}{ccc} E_f & \longrightarrow & B^I \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array} \quad \begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A \times I & \longrightarrow & M_f \end{array}$$

Thus the pushout is a quotient of $X \amalg Y$, while the pullback of maps $X \rightarrow Z$ and $Y \rightarrow Z$ is a subobject of $X \times Y$, so we see here two instances of duality: a duality between disjoint union and product, and a duality between subobjects and quotients. The first of these is easily explained, since a collection of maps $X_\alpha \rightarrow X$ is equivalent to a map $\coprod_\alpha X_\alpha \rightarrow X$, while a collection of maps $X \rightarrow X_\alpha$ is equivalent to a map $X \rightarrow \prod_\alpha X_\alpha$. The notation $\coprod$ for the ‘coproduct’ was chosen to indicate that it is dual to $\prod$. If we were dealing with basepointed spaces and maps, the coproduct would be wedge sum. In the category of abelian groups the coproduct is direct sum.

The duality between subobjects and quotient objects is clear for abelian groups, where subobjects are kernels and quotient objects cokernels. The strict topological analog of a kernel is a fiber of a fibration. Dually, the topological analog of a cokernel is the **cofiber** B/A of a cofibration $A \hookrightarrow B$. If we make an arbitrary map $f : A \rightarrow B$ into a cofibration $A \hookrightarrow M_f$, the cofiber is the mapping cone $C_f = M_f / (A \times \{0\})$.

In the diagram showing E_f and M_f as pullback and pushout, there also appears to be some sort of duality involving the terms $A \times I$ and B^I . This leads us to ask whether $X \times I$ and X^I are in some way dual. Indeed, if we ignore topology and just think set-theoretically, this is an instance of the familiar product-coproduct duality since the product of copies of X indexed by I is X^I , all functions $I \rightarrow X$, while the coproduct of copies of X indexed by I is $X \times I$, the disjoint union of the sets $X \times \{t\}$ for $t \in I$. Switching back from the category of sets to the topological category, we can view X^I as a ‘continuous product’ of copies of X and $X \times I$ as a ‘continuous coproduct.’

On a less abstract level, the fact that maps $A \times I \rightarrow B$ are the same as maps $A \rightarrow B^I$ indicates a certain duality between $A \times I$ and B^I . This leads to a duality between